

DDIA Lab 3 Report

Team Members

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Deliverables

- A running version of the cassandra cluster on docker
- Screenshots showing terminal output in each of the cases

From the Start

```
C:\Users\PC\Documents\2nd term 3rd year\DDIA\Labs\Lab 3>wsl
root@DESKTOP-MJ53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# sudo systemctl start docker
root@DESKTOP-MJ53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# sudo systemctl enable docker
sudo: systemctl: command not found
root@DESKTOP-MJ53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# sudo systemctl enable docker
root@DESKTOP-MJ53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker --version
Docker version 26.1.3, build 26.1.3-0ubuntu1-24.04.1
root@DESKTOP-MJ53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker-compose up -d
WARN[0000] /mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[*] Running 13/13
  [+] cassandra2 Pulled
  [+] cassandra-seed Pulled
  [+] 9cb31e2e37ee Pull complete
  [+] 13876c9bdc5 Pull complete
  [+] 25f4fc9faee8 Pull complete
  [+] b682cc54ed35 Pull complete
  [+] 4615ed3a3407 Pull complete
  [+] e4812c6119a3 Pull complete
  [+] a98a9ced27c2 Pull complete
  [+] 39ba2eb345fd Pull complete
  [+] 61ae76b66881 Pull complete
  [+] 0672c4590eb9 Pull complete
  [+] cassandra1 Pulled
[*] Running 2/7
  [+] Network lab3_app-tier Created
  [+] Volume "lab3_cassandra_data_2" Created
  [+] Volume "lab3_cassandra_data_seed" Created
  [+] Volume "lab3_cassandra_data_1" Created
  [+] Container lab3-cassandra-seed-1 Started
  [+] Container lab3-cassandra-2-1 Started
  [+] Container lab3-cassandra-1 Started
root@DESKTOP-MJ53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# sudo docker run hello-world
```

```
root@DESKTOP-MJ53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED      STATUS      PORTS
c226161e5152   cassandra:latest   "docker-entrypoint.s..." 6 minutes ago Up 6 minutes 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, :::9041->9042/tcp   lab3-cassandra-1
319b6dc6bf4    cassandra:latest   "docker-entrypoint.s..." 6 minutes ago Up 6 minutes 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, :::9040->9042/tcp   lab3-cassandra-2-1
e2e89272b0d2   cassandra:latest   "docker-entrypoint.s..." 6 minutes ago Up 6 minutes 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp   lab3-cassandra-seed-1
root@DESKTOP-MJ53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker exec -it lab3-cassandra-seed-1 cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.3 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> SELECT * FROM system.peers;

peer           | data_center | host_id | preferred_ip | rack | release_version | ip_address | schema_version | tokens
-----
172.18.0.4 |      sel     | e225643b-c17a-4d6e-a7fa-3d078f5a7ea0 | null | rack1 | 5.0.3 | 172.18.0.4 | 003783d7-b408-3c1a-02f1-0e9052ed6d0b |
172.18.0.3 |      sel     | 3c40b7aa-e672-45b4-9581-71c201a3979d | null | rack1 | 5.0.3 | 172.18.0.3 | 003783d7-b408-3c1a-02f1-0e9052ed6d0b | {'-1338075137507281212', '-2384416127280344008', '-3652816718423997642', '-574531618426232752', '-6831988135355198969', '-818210996192988853', '-80803964631210998243', '-2366993492985651214', '-3535948416164665593', '-4585072216493856681', '-489929163472665185', '-5520965389285372181', '-5936623048857011901', '-7148303395328993126', '-754362624777795917', '-8084840363023149003'}
(2 rows)
cqlsh> CREATE KEYSPACE bookstore WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
CREATE TABLE books (
  name VARCHAR,
  category VARCHAR,
  year INT,
  title VARCHAR,
  PRIMARY KEY ((name), category, year, title)
) WITH CLUSTERING ORDER BY (category ASC, year DESC, title ASC);
cqlsh> USE bookstore;
cqlsh> CREATE TABLE books (
  ... name VARCHAR,
  ... category VARCHAR,
  ... year INT,
  ... title VARCHAR,
  ... PRIMARY KEY ((name), category, year, title)
  ... ) WITH CLUSTERING ORDER BY (category ASC, year DESC, title ASC);
cqlsh> INSERT INTO books (name, category, year, title) VALUES ('BookA', 'Fiction', 2023, 'Story A');
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';

name | category | year | title
-----
BookA | Fiction | 2023 | Story A
```

Mat 1 RF = 1 (Data was stored only on cas1)

when cas 1 is paused query didn't work

```
(3 rows)
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';
ReadTimeout: Error from server: code=1200 [Coordinator node timed out waiting for replica nodes' responses] message="Operation timed out - received only 0 responses." info={'consistency': 'ONE', 'required_responses': 1, 'received_responses': 0}
cqlsh> bookstore

[2025-03-21 11:27:03.538] Replication factor is 1. No repair is needed for keyspace 'bookstore'
root@DESKTOP-MJ53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container pause lab3-cassandra-1
root@DESKTOP-MJ53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#
```

when cas 2 paused query worked

```
(1 rows)
cqlsh:bookstore> SELECT * FROM bookstore.books WHERE name='BookA';
readtimeout: Error from server: code=1208 [Coordinator node timed out waiting for replica nodes' responses] message="Operation timed out - received only 0 responses." info={'consistency': 'ONE', 'required_responses': 1, 'received_responses': 0}
cqlsh:bookstore> SELECT * FROM bookstore.books WHERE name='BookA';

name | category | year | title
-----
BookA | Fiction | 2023 | Story A

(1 rows)
cqlsh:bookstore> SELECT * FROM bookstore.books WHERE name='BookA';

name | category | year | title
-----
BookA | Fiction | 2023 | Story A

(1 rows)
cqlsh:bookstore>
```

```
[2025-03-21 11:27:03.530] Replication factor is 1. No repair is needed for keyspace 'bookstore'
root@DESKTOP-MU530P6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DOIA/Labs/Lab 3# docker container pause lab3-cassandra-1
lab3-cassandra-1
root@DESKTOP-MU530P6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DOIA/Labs/Lab 3# docker container unpause lab3-cassandra-1
lab3-cassandra-1
root@DESKTOP-MU530P6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DOIA/Labs/Lab 3# docker container pause lab3-cassandra-1
lab3-cassandra-1
root@DESKTOP-MU530P6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DOIA/Labs/Lab 3#
```

when seed is paused query worked

```
(1 rows)
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';

name | category | year | title
-----
BookA | Fiction | 2023 | Story A

(1 rows)
cqlsh>
```

```
9/tcp, 9100/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp lab3-cassandra-seed-1
root@DESKTOP-MU530P6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DOIA/Labs/Lab 3# docker container pause lab3-cassandra-seed-1
lab3-cassandra-seed-1
root@DESKTOP-MU530P6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DOIA/Labs/Lab 3#
```

when cas 1, 2 are paused query didn't work

```
cqlsh:bookstore> SELECT * FROM bookstore.books WHERE name='BookA';
readtimeout: Error from server: code=1208 [Coordinator node timed out waiting for replica nodes' responses] message="Operation timed out - received only 0 responses." info={'consistency': 'ONE', 'required_responses': 1, 'received_responses': 0}
cqlsh:bookstore>
```

```
lab3-cassandra-1
lab3-cassandra-1
root@DESKTOP-MU530P6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DOIA/Labs/Lab 3# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
c2763fe5152   cassandra:late "docker-entrypoint.s..." 4 hours ago   Up 4 hours   (Paused)    7000-7001
/tcp, 7199/tcp, 9100/tcp, 0.0.0.0:9041->9042/tcp, :::9041->9042/tcp lab3-cassandra-1
110b6fdecd44   cassandra:late "docker-entrypoint.s..." 4 hours ago   Up 4 hours   (Paused)    7000-7001
/tcp, 7199/tcp, 9100/tcp, 0.0.0.0:9040->9042/tcp, :::9040->9042/tcp lab3-cassandra-2-1
e2b092720bd2   cassandra:late "docker-entrypoint.s..." 4 hours ago   Up 4 hours   7000-7001
/tcp, 7199/tcp, 9100/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp lab3-cassandra-seed-1
root@DESKTOP-MU530P6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DOIA/Labs/Lab 3#
```

when cas 1 and seed are paused query didn't work

```
(0 rows)
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';

name | category | year | title
-----
(0 rows)
cqlsh>
```

```
lab3-cassandra-2-1
lab3-cassandra-2-1
root@DESKTOP-PJ5JDP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DOIA/Labs/Lab 3# docker container pause lab3-cassandra-1 lab3-cassandra-seed-1
lab3-cassandra-1
lab3-cassandra-seed-1
root@DESKTOP-PJ5JDP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DOIA/Labs/Lab 3#
```

when cas 2 and seed are paused query worked

```
(1 rows)
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';

name | category | year | title
-----
BookA | Fiction | 2021 | Story A
(1 rows)
cqlsh>
```

```
lab3-cassandra-seed-1
lab3-cassandra-seed-1
lab3-cassandra-seed-1
lab3-cassandra-seed-1
root@DESKTOP-PJ5JDP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DOIA/Labs/Lab 3#
```

data is **stored only on cas 1** and we cant do anything about it since consistency is already ONE

RF = 2, Consistency = QUORUM (Data was stored on cas1 and seed)

consistency quorum (rf/2 + 1) requires that 2 out of 3 nodes to be alive

```
cqlsh:bookstore> ALTER KEYSPACE bookstore WITH REPLICATION = { 'class' : 'SimpleStrategy', 'replication_factor' : 2 };

Warnings :
When increasing replication factor you need to run a full (-full) repair to distribute the data.

cqlsh:bookstore> CONSISTENCY QUORUM;
consistency level set to QUORUM.
cqlsh:bookstore>
```

```
lab3-cassandra-seed-1
root@DESKTOP-PJ5JDP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DOIA/Labs/Lab 3# docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED         STATUS         PORTS
c2763f3e5152   cassandra:latest   "docker-entrypoint.s..." 4 hours ago    Up 4 hours     7000-7001/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, :::9041->9042/tcp
1f86ede6cbf4   cassandra:latest   "docker-entrypoint.s..." 4 hours ago    Up 4 hours     7000-7001/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, :::9040->9042/tcp
e2e89272b0d2   cassandra:latest   "docker-entrypoint.s..." 4 hours ago    Up 4 hours     7000-7001/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp
lab3-cassandra-seed-1
root@DESKTOP-PJ5JDP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DOIA/Labs/Lab 3# docker exec -it lab3-cassandra-seed-1 node tool repair -full bookstore
[2025-03-21 12:52:27,812] Starting repair command #3 (58118200-0653-11f0-8c57-5f533d1b476f), repairing keyspace bookstore with repair options (parallelism: parallel, primary range: false, incremental: false, job threads: 1, ColumnFamilies: [], dataCenters: [], hosts: [], previousKind: NONE, # of ranges: 32, pull repair: false, force repair: false, optimise streams: false, ignore unreplicated keyspaces: false, repairSession: true, passOnly: false)
[2025-03-21 12:52:27,868] Repair session 58118200-0653-11f0-8c57-5f533d1b476f for range [(17862983684487, 39115, 2166993692985651214), (6586559977169366882, 7148303395528993126), (-681190813555108969, -638813914264561943), (-6188139147268501043, -5774531618426232752), (9183728725382524255, -8688196463210998243), (-1940896542842511874, -1338875113787281212), (2366993482985651214, 2785671175896348771), (-8182189961929868853, -7484381847583448212), (2785671175896348771, 953948416164665593), (-923845138108844758, -10537320548774884), (-4452665320685127926, -3652818718423997642), (4996355424855466244, 592008339928372181), (880484663821448895, 9183728725382524255), (-10537320548774884, 489929163472665185), (4829191181576932245, 4389872216493856681), (7148303395528993126, 7545626247877705917), (-8688396463210998243, -8182189961929868853), (-3652818718423997642, -3012397359182544686), (489929163472665185, 970382675737845474), (5936623048857011901, 6586559977169366882), (-1338875113787281212, 923845138108844758), (7545626247877705917, 820558127286848179), (-7484381847583448212, -681398115395398993), (-239441612728034468, 158896542842511874), (4389872216493856681, 4996555424855466244), (-52978420346578289683, -4452665389885127926), (-5774531618426232752, -52978420346578289683), (820558127286848179, 880484663821448895), (3535948416164665593, 4829191181576932245), (5520885389285372181, 5936623048857011981), (-3012397359182544686, -2394416127280344688), (970382675737845474, 1786298368448739115)] finished (progress: 57%)
[2025-03-21 12:52:27,872] Repair completed successfully
[2025-03-21 12:52:27,872] Repair command #3 finished in 0 seconds.
[2025-03-21 12:52:27,875] condition satisfied queried for parent session status and discovered repair completed.
[2025-03-21 12:52:27,875] Repair completed successfully
root@DESKTOP-PJ5JDP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DOIA/Labs/Lab 3#
```

when cas 1 is paused query didn't work

```
(1 rows)
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';
ReadTimeout: Error from server: code=1200 [Coordinator node timed out waiting for replica nodes' responses] message="Operation timed out - received only 1 responses." info={'consistency': 'QUORUM', 'required_responses': 2, 'received_responses': 1}
cqlsh> bookstore>
```

```
Completed.
[2025-03-21 12:52:27.875] Repair completed successfully
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container pause lab3-cassandra1-1
lab3-cassandra1-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#
```

When cas 1 is paused but consistency set to one query worked

```
cqlsh> CONSISTENCY QUORUM;
Consistency level set to QUORUM.
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';
Unavailable: ('Unable to complete the operation against any hosts', (Host: 127.0.0.1:9042 set: Unavailable('Error from server: code=1000 [Unavailable exception] message="Cannot achieve consistency level QUORUM" info={'consistency': 'QUORUM', 'required_replicas': 2, 'alive_replicas': 1}'))))
cqlsh> CONSISTENCY ONE;
Consistency level set to ONE.
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';

name | category | year | title
-----
BookA | Fiction | 2023 | Story A
(1 rows)
cqlsh>
```

```
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
c2763f3e5152   cassandra:latest "docker-entrypoint.s..." 14 hours ago   Up 14 hours   7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, :::9041->9042/tcp   lab3-cassandra1-1
319be6decbf4   cassandra:latest "docker-entrypoint.s..." 14 hours ago   Up 14 hours   7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, :::9040->9042/tcp   lab3-cassandra2-1
e2e892720bd2   cassandra:latest "docker-entrypoint.s..." 14 hours ago   Up 14 hours (Paused) 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp   lab3-cassandra-seed-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#
```

when cas 2 is paused query worked

```
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';

name | category | year | title
-----
BookA | Fiction | 2023 | Story A
(1 rows)
cqlsh>
```

```
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container pause lab3-cassandra1-1
lab3-cassandra1-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container unpause lab3-cassandra1-1
lab3-cassandra1-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container pause lab3-cassandra2-1
lab3-cassandra2-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#
```

When seed is paused query didn't work

```
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';
ReadTimeout: Error from server: code=1200 [Coordinator node timed out waiting for replica nodes' responses] message="Operation timed out - received only 1 responses." info={'consistency': 'QUORUM', 'required_responses': 2, 'received_responses': 1}
cqlsh>
```

```
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
c2763f3e5152   cassandra:latest "docker-entrypoint.s..." 14 hours ago   Up 14 hours   7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, :::9041->9042/tcp   lab3-cassandra1-1
319be6decbf4   cassandra:latest "docker-entrypoint.s..." 14 hours ago   Up 14 hours   7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, :::9040->9042/tcp   lab3-cassandra2-1
e2e892720bd2   cassandra:latest "docker-entrypoint.s..." 14 hours ago   Up 14 hours (Paused) 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp   lab3-cassandra-seed-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#
```

When seed is paused but consistency set to one query worked

```
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';
ReadTimeout: Error from server: code=1200 [Coordinator node timed out waiting for replica nodes' responses] message="Operation timed out - received only 1 responses." info={'consistency': 'QUORUM', 'required_replicas': 2, 'received_responses': 1}
cqlsh> CONSISTENCY ONE;
Consistency level set to ONE.
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';
```

name	category	year	title
BookA	Fiction	2023	Story A

```
(1 rows)
cqlsh>
```

```
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container pause lab3-cassandra-seed-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	NAMES	PORTS
c2763f3e5152	cassandra:latest	"docker-entrypoint.s..."	14 hours ago	Up 14 hours	lab3-cassandra-1	7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, :::9041->9042/tcp
11b06decbf4	cassandra:latest	"docker-entrypoint.s..."	14 hours ago	Up 14 hours	lab3-cassandra-2-1	7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, :::9040->9042/tcp
e2e89272b0d2	cassandra:latest	"docker-entrypoint.s..."	14 hours ago	Up 14 hours (Paused)	lab3-cassandra-seed-1	7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp

```
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#
```

when both cas 1, 2 are paused query didn't work

```
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';
ReadTimeout: Error from server: code=1200 [Coordinator node timed out waiting for replica nodes' responses] message="Operation timed out - received only 1 responses." info={'consistency': 'QUORUM', 'required_replicas': 2, 'received_responses': 1}
cqlsh>
```

```
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container pause lab3-cassandra-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#
```

when both cas 1, 2 are paused but consistency set to ONE query worked

```
cqlsh> CONSISTENCY QUORUM;
Consistency level set to QUORUM.
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';
HostsAvailable: {'Unable to complete the operation against any hosts', '{(Host: 127.0.0.1:9042 sel): Unavailable('Error from server: code=1000 [Unavailable exception] message="Cannot achieve consistency level QUORUM" info={'consistency': 'QUORUM', 'required_replicas': 2, 'alive_replicas': 1})}'}
```

name	category	year	title
BookA	Fiction	2023	Story A

```
cqlsh>
```

```
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container pause lab3-cassandra-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#
```

when both cas 1 and seed are paused query didn't work since required replicas = 2 alive replicas = 1

```
cqlsh> CONSISTENCY ONE;
Current consistency level is QUORUM.
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';
HostsAvailable: {'Unable to complete the operation against any hosts', '{(Host: 127.0.0.1:9042 sel): Unavailable('Error from server: code=1000 [Unavailable exception] message="Cannot achieve consistency level QUORUM" info={'consistency': 'QUORUM', 'required_replicas': 2, 'alive_replicas': 1})}'}
```

```
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container pause lab3-cassandra-seed-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#
```

when both cas 1 and seed are paused and consistency set to ONE query worked but returned 0 rows indicating that data is stored in **cas 1 and seed**

```

cqlsh> CONSISTENCY;
Current consistency level is QUORUM.
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';
Unavailable: ('Unable to complete the operation against any hosts', {'host: 127.0.0.1:9042 sel: Unavailable', 'Error from server: code=1000 [Unavailable exception] message="cannot achieve consistency level QUORUM" info={\'consistency\': \'QUORUM\', \'required_replicas\': 2, \'alive_replicas\': 1}})})
cqlsh> CONSISTENCY ONE;
Consistency level set to ONE.
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';

name | category | year | title
-----
(0 rows)
cqlsh>

```

```

se lab3-cassandra-1
lab3-cassandra-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container pause lab3-cassandra-seed-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS         PORTS
c2763f3e5152   cassandra:latest "docker-entrypoint.s..." 9 hours ago    Up 9 hours (Paused) 7000-7001
01/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, :::9041->9042/tcp   lab3-cassandra-1
319b6dec6cf4   cassandra:latest "docker-entrypoint.s..." 9 hours ago    Up 9 hours      7000-7001
01/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, :::9040->9042/tcp   lab3-cassandra-2-1
e2e892728bd2   cassandra:latest "docker-entrypoint.s..." 9 hours ago    Up 9 hours (Paused) 7000-7001
01/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp   lab3-cassandra-seed-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#

```

when both cas 2 and seed are paused query didn't work since required replicas = 2 alive replicas = 1

```

cqlsh> CONSISTENCY;
Current consistency level is QUORUM.
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';
Unavailable: ('Unable to complete the operation against any hosts', {'host: 127.0.0.1:9042 sel: Unavailable', 'Error from server: code=1000 [Unavailable exception] message="cannot achieve consistency level QUORUM" info={\'consistency\': \'QUORUM\', \'required_replicas\': 2, \'alive_replicas\': 1}})})
cqlsh>

```

```

root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS         PORTS
c2763f3e5152   cassandra:latest "docker-entrypoint.s..." 10 hours ago    Up 10 hours      7000-7001
01/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, :::9041->9042/tcp   lab3-cassandra-1
319b6dec6cf4   cassandra:latest "docker-entrypoint.s..." 10 hours ago    Up 10 hours (Paused) 7000-7001
01/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, :::9040->9042/tcp   lab3-cassandra-2-1
e2e892728bd2   cassandra:latest "docker-entrypoint.s..." 10 hours ago    Up 10 hours (Paused) 7000-7001
01/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp   lab3-cassandra-seed-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#

```

when both cas 2 and seed are paused but consistency set to ONE query worked

```

cqlsh> CONSISTENCY;
Current consistency level is QUORUM.
cqlsh> CONSISTENCY;
Current consistency level is QUORUM.
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';
Unavailable: ('Unable to complete the operation against any hosts', {'host: 127.0.0.1:9042 sel: Unavailable', 'Error from server: code=1000 [Unavailable exception] message="cannot achieve consistency level QUORUM" info={\'consistency\': \'QUORUM\', \'required_replicas\': 2, \'alive_replicas\': 1}})})
cqlsh> CONSISTENCY ONE;
Consistency level set to ONE.
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';

name | category | year | title
-----
BookA | Fiction | 2023 | Story A
(1 rows)
cqlsh>

```

```

lab3-cassandra-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS         PORTS
c2763f3e5152   cassandra:latest "docker-entrypoint.s..." 10 hours ago    Up 10 hours      7000-7001
01/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, :::9041->9042/tcp   lab3-cassandra-1
319b6dec6cf4   cassandra:latest "docker-entrypoint.s..." 10 hours ago    Up 10 hours (Paused) 7000-7001
01/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, :::9040->9042/tcp   lab3-cassandra-2-1
e2e892728bd2   cassandra:latest "docker-entrypoint.s..." 10 hours ago    Up 10 hours (Paused) 7000-7001
01/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp   lab3-cassandra-seed-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#

```

RF = 3 and Consistency = QUORUM

when cas 1 is paused query worked

```

cqlsh> CONSISTENCY QUORUM;
Consistency level set to QUORUM.
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';

name | category | year | title
-----
BookA | Fiction | 2023 | Story A
(1 rows)
cqlsh> CONSISTENCY;
Current consistency level is QUORUM.
cqlsh>

```

```

root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container pause lab3-cassandra-1
lab3-cassandra-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#

```

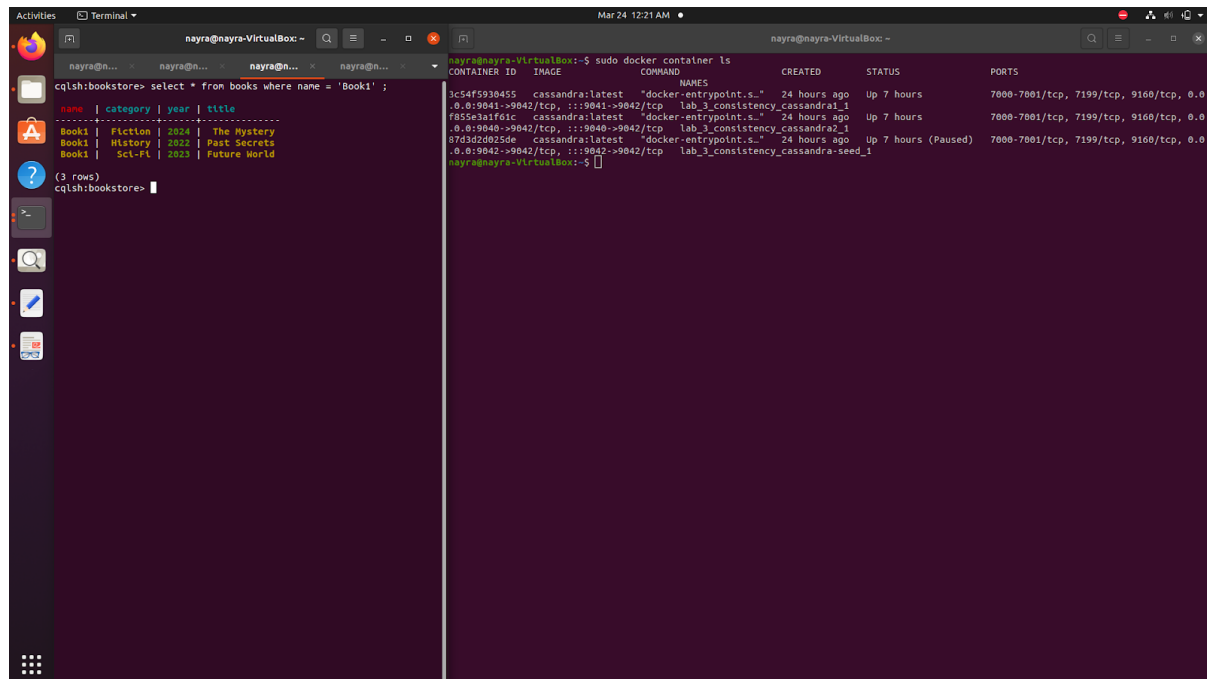
when cas 2 is paused query worked

```
qlsh> CONSISTENCY;
current consistency level is QUORUM.
qlsh> SELECT * FROM bookstore.books WHERE name='BookA';

name | category | year | title
-----
BookA | Fiction | 2023 | Story A
(1 rows)
qlsh>
```

```
root@DESKTOP-MU530P6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container unpau
lab3-cassandra1-1
lab3-cassandra1-1
root@DESKTOP-MU530P6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container pause
lab3-cassandra2-1
lab3-cassandra2-1
root@DESKTOP-MU530P6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#
```

when seed is paused query worked



The screenshot shows a terminal window with a table of books. The query 'select * from books where name = 'Book1';' has been executed, returning three rows of data. The table has columns: name, category, year, and title.

name	category	year	title
Book1	Fiction	2024	The Mystery
Book1	History	2022	Past Secrets
Book1	Sci-Fi	2023	Future World

when both cas 1 , 2 are paused query didn't work

```
qlsh:bookstore> SELECT * FROM bookstore.books WHERE name='BookA';
readtimeout: Error from server: code=1200 [Coordinator node timed out waiting for replica nodes' responses] message="Operation timed out - received only 1 responses." info={'consistency': 'QUORUM', 'required_responses': 2, 'received_responses': 1}
qlsh:bookstore>
```

```
root@DESKTOP-MU530P6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container pause
lab3-cassandra2-1 lab3-cassandra1-1
lab3-cassandra2-1
lab3-cassandra1-1
root@DESKTOP-MU530P6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#
```

when both cas 1 , 2 are paused but consistency set to ONE query worked

```
cqlsh:bookstore> SELECT * FROM bookstore.books WHERE name='BookA';
ReadTimeout: Error from server: code=1200 [Coordinator node timed out waiting for replica nodes' responses] message="Operation timed out - received only 1 responses." info={'consistency': 'QUORUM', 'required_responses': 2, 'received_responses': 1}
cqlsh:bookstore> CONSISTENCY ONE;
Consistency level set to ONE.
cqlsh:bookstore> SELECT * FROM bookstore.books WHERE name='BookA';

name | category | year | title
-----
BookA | Fiction | 2023 | Story A
(1 rows)
cqlsh:bookstore>
```

```
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container pause lab3-cassandra2-1 lab3-cassandra1-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#
```

when both cas 1 and seed were paused query didn't work

```
(1 rows)
cqlsh> CONSISTENCY QUORUM;
outofsequence: line 1:0 no viable alternative at input 'CONSISTENCYQUORUM' ([CONSISTENCYQUORUM]...)
cqlsh> CONSISTENCY QUORUM;
Consistency level set to QUORUM.
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';
ReadTimeout: Error from server: code=1200 [Coordinator node timed out waiting for replica nodes' responses] message="Operation timed out - received only 1 responses." info={'consistency': 'QUORUM', 'required_responses': 2, 'received_responses': 1}
cqlsh>
```

```
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container unpause lab3-cassandra2-1 lab3-cassandra1-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container pause lab3-cassandra-2-1 lab3-cassandra-1-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#
```

when both cas 1 and seed were paused but consistency set to one query worked

```
ReadTimeout: Error from server: code=1200 [Coordinator node timed out waiting for replica nodes' responses] message="Operation timed out - received only 1 responses." info={'consistency': 'QUORUM', 'required_responses': 2, 'received_responses': 1}
cqlsh> CONSISTENCY ONE;
Consistency level set to ONE.
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';

name | category | year | title
-----
BookA | Fiction | 2023 | Story A
(1 rows)
cqlsh>
```

```
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container unpause lab3-cassandra2-1 lab3-cassandra1-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker container pause lab3-cassandra-2-1 lab3-cassandra-1-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#
```

when both cas 2 and seed were paused query didn't work

```
cqlsh> CONSISTENCY QUORUM;
Consistency level set to QUORUM.
cqlsh> SELECT * FROM bookstore.books WHERE name='BookA';
NotHostAvailable: ('Unable to complete the operation against any hosts', {'Host: 127.0.0.1:9042 (seed): Unavailable', 'Error from server: code=1000 [Unavailable exception] message="Cannot achieve consistency level QUORUM" info={'consistency': 'QUORUM', 'required_replicas': 2, 'alive_replicas': 1}})})
cqlsh>
```

```
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
c2763f3e5152   cassandra:latest                  "docker-entrypoint.s..." 13 hours ago   Up 13 hours   7000-70
01/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, ::9041->9042/tcp   lab3-cassandra1-1
3198e6decbf4   cassandra:latest                  "docker-entrypoint.s..." 13 hours ago   Up 13 hours   7000-70
01/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, ::9040->9042/tcp   lab3-cassandra2-1
e26892728bd2   cassandra:latest                  "docker-entrypoint.s..." 13 hours ago   Up 13 hours   7000-70
01/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, ::9042->9042/tcp   lab3-cassandra-seed-1
root@DESKTOP-MU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/DDIA/Labs/Lab 3#
```

when both cas 2 and seed were paused but consistency set to one query worked

```
consistency level set to QUORUM.
qlsh> SELECT * FROM bookstore.books WHERE name='BookA';
Unselectable: ('Unable to complete the operation against any hosts', {dbhost: 127.0.0.1:9042 sel): Unavailable('Error from server: code=1000 (Unavailable exception) message="cannot achieve consistency level QUORUM" info={\'consistency\': \'QUORUM\', \'required_replicas\': 2, \'alive_replicas\': 1}})')
qlsh> CONSISTENCY ONE;
consistency level set to ONE.
qlsh> SELECT * FROM bookstore.books WHERE name='BookA';

name | category | year | title
-----|-----|-----|-----
BookA | Fiction | 2023 | Story A
(1 rows)
qlsh>
```

```
se lab3-cassandra-1
lab3-cassandra-1
root@DESKTOP-RU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/001A/Labs/Lab 3# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS
c2763f3e5152   cassandra:latest   "docker-entrypoint.s..." 13 hours ago    Up 13 hours    7000-7001/tcp, 9160/tcp, 0.0.0.0:9041-9042/tcp, :::9041-9042/tcp   lab3-cassandra-1
110b0dc1b4f4   cassandra:latest   "docker-entrypoint.s..." 13 hours ago    Up 13 hours (Paused)    7000-7001/tcp, 9160/tcp, 0.0.0.0:9040-9042/tcp, :::9040-9042/tcp   lab3-cassandra-2.1
c2e892720bd2   cassandra:latest   "docker-entrypoint.s..." 13 hours ago    Up 13 hours (Paused)    7000-7001/tcp, 9160/tcp, 0.0.0.0:9042-9042/tcp, :::9042-9042/tcp   lab3-cassandra-seed-1
vnot8DESKTOP-RU53DP6:/mnt/c/Users/PC/Documents/2nd term 3rd year/001A/Labs/Lab 3#
```

- Table in requirement D

Nodes Down	RF = 1	RF = 2	RF = 3
Cas 1	Failure	Failure	Success
Cas 2	Success	Success	Success
Seed	Success	Failure	Success
Cas 1 & 2	Failure	Failure	Failure
Cas 1 & Seed	Failure	Failure	Failure
Cas 2 & Seed	Success	Failure	Failure

Visualization:

Cassandra Node Availability (Success = Green, Failure = Red)				
Nodes Down	Cas 1	0	0	1
	Cas 2	1	1	1
	Seed	1	0	1
	Cas 1 & 2	0	0	0
	Cas 1 & Seed	0	0	0
	Cas 2 & Seed	1	0	0
		RF = 1	RF = 2	RF = 3
		Replication Factor		

Nodes Representation:

Node1 : Cassandra1-1

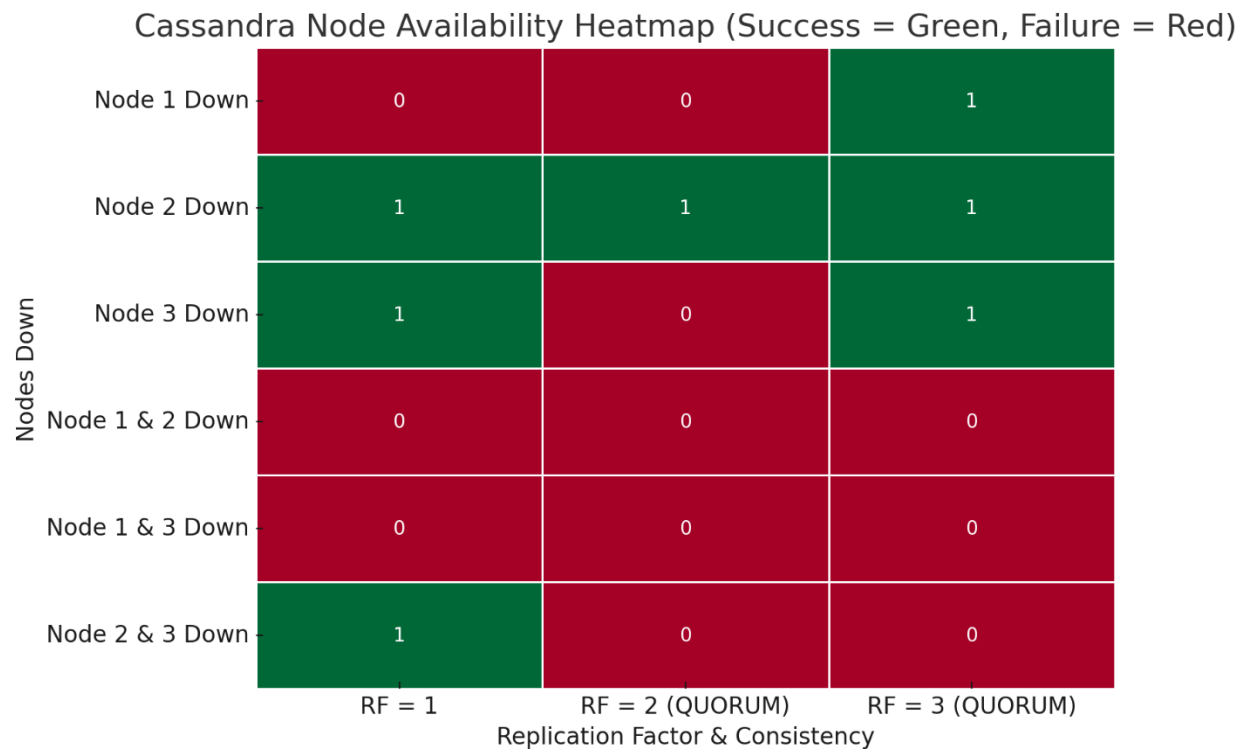
Node2 : Cassandra2-1

Node3 : Cassandra-seed-1

Insertion (write) summary

Nodes Down	RF = 1	RF = 2 (QUORUM)	RF = 3 (QUORUM)
Node 1 down	ERROR	ERROR	OK
Node 2 down	OK	OK	OK
Node 3 down	OK	ERROR	OK
Node 1&2 down	ERROR	ERROR	ERROR
Node 1&3 down	ERROR	ERROR	ERROR
Node 2&3 down	OK	ERROR	ERROR

Visualization



RF = 1 , Node 1 down

```
Activities Terminal Mar 23 9:33 PM
nayra@naya-VirtualBox -
nayra@naya-VirtualBox -

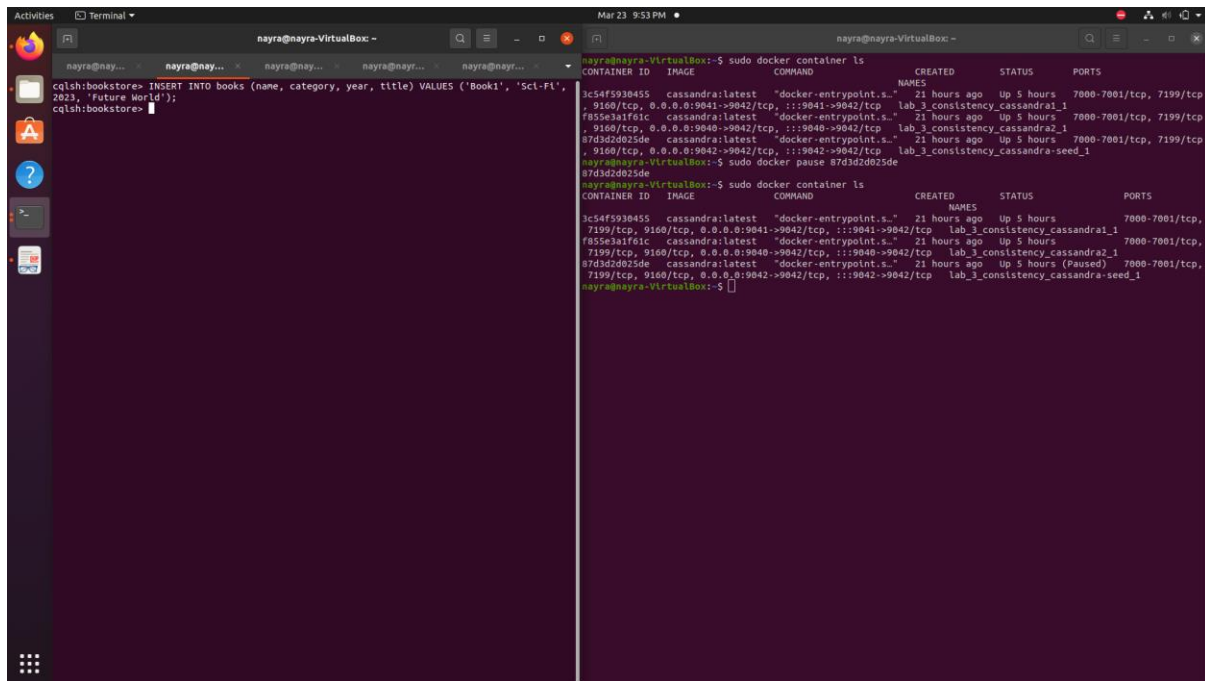
nayra@nay... x nayra@nay... x nayra@nay... x nayra@nay... x
cqlsh:bookstore> INSERT INTO books (name, category, year, title) VALUES ('Book1', 'Fiction',
2024, 'The Mystery');
WriteInnovati: Error from server: code=1100 [Coordinator node timed out waiting for replica n
node responses] Message: Operation timed out: received only 0 responses.: info=[localhost
cs:1] [0ms] [required_responses:1, received_responses:0, write_type: 'sync']
cqlsh:bookstore>

nayra@naya-VirtualBox:~$ sudo docker container ls
[sudo] password for nayra:
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
3c54f5930455 cassandra:latest "docker-entrypoint.s..." 21 hours ago Up 4 hours 7000-7001/tcp, 7199/tcp
9160/tcp, 0.0.0.0:9041->9042/tcp, ::9041->9042/tcp lab_3_consistency_cassandra1_1
f855e3a1f01c cassandra:latest "docker-entrypoint.s..." 21 hours ago Up 4 hours 7000-7001/tcp, 7199/tcp
9160/tcp, 0.0.0.0:9040->9042/tcp, ::9040->9042/tcp lab_3_consistency_cassandra2_1
8fd3d2d025de cassandra:latest "docker-entrypoint.s..." 21 hours ago Up 4 hours 7000-7001/tcp, 7199/tcp
9160/tcp, 0.0.0.0:9042->9042/tcp, ::9042->9042/tcp lab_3_consistency_cassandra-seed_1
nayra@naya-VirtualBox:~$ sudo docker pause 3c54f5930455
3c54f5930455
nayra@naya-VirtualBox:~$ sudo docker container ls
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
3c54f5930455 cassandra:latest "docker-entrypoint.s..." 21 hours ago Up 4 hours (Paused) 7000-7001/tcp,
7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, ::9041->9042/tcp lab_3_consistency_cassandra1_1
f855e3a1f01c cassandra:latest "docker-entrypoint.s..." 21 hours ago Up 4 hours 7000-7001/tcp,
7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, ::9040->9042/tcp lab_3_consistency_cassandra2_1
8fd3d2d025de cassandra:latest "docker-entrypoint.s..." 21 hours ago Up 4 hours 7000-7001/tcp,
7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, ::9042->9042/tcp lab_3_consistency_cassandra-seed_1
nayra@naya-VirtualBox:~$
```

RF = 1 , Node 2 down

[illegible]

RF = 1 , Node 3 down



The screenshot shows a terminal window with the following content:

```
nayra@nayra-VirtualBox:~$ sudo docker container ls
CONTAINER ID        IMAGE               COMMAND                  CREATED             STATUS              PORTS
3c54f5938455        cassandra:latest   "docker-entrypoint.s..." 21 hours ago        Up 5 hours          7000-7001/tcp, 7199/tcp
, 9160/tcp, 0.0.0.0:9041->9042/tcp, :::9041->9042/tcp    lab_3_consistency_cassandra1_1
f855e3a1f61c        cassandra:latest   "docker-entrypoint.s..." 21 hours ago        Up 5 hours          7000-7001/tcp, 7199/tcp
, 9160/tcp, 0.0.0.0:9040->9042/tcp, :::9040->9042/tcp    lab_3_consistency_cassandra2_1
87d3d2d025de        cassandra:latest   "docker-entrypoint.s..." 21 hours ago        Up 5 hours          7000-7001/tcp, 7199/tcp
, 9160/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp    lab_3_consistency_cassandra-seed_1
nayra@nayra-VirtualBox:~$ sudo docker pause 87d3d2d025de
87d3d2d025de
nayra@nayra-VirtualBox:~$ sudo docker container ls
CONTAINER ID        IMAGE               COMMAND                  CREATED             STATUS              PORTS
3c54f5938455        cassandra:latest   "docker-entrypoint.s..." 21 hours ago        Up 5 hours          7000-7001/tcp,
7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, :::9041->9042/tcp    lab_3_consistency_cassandra1_1
f855e3a1f61c        cassandra:latest   "docker-entrypoint.s..." 21 hours ago        Up 5 hours          7000-7001/tcp,
7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, :::9040->9042/tcp    lab_3_consistency_cassandra2_1
87d3d2d025de        cassandra:latest   "docker-entrypoint.s..." 21 hours ago        Up 5 hours (Paused) 7000-7001/tcp,
7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp    lab_3_consistency_cassandra-seed_1
nayra@nayra-VirtualBox:~$
```

In the background, a web browser window is open showing a SQL query result for a 'bookstore' database. The query is: `INSERT INTO books (name, category, year, title) VALUES ('Book1', 'Sci-Fi', 2023, 'Future World');`. The result shows a single row with the values: `Book1`, `Sci-Fi`, `2023`, `Future World`.

RF = 1 , Node 1&2 down

```
Activities Terminal Mar 23 9:55 PM
nayra@nayra-VirtualBox:~$ sudo docker container ls
CONTAINER ID        IMAGE               COMMAND                  CREATED        STATUS        PORTS
3c54f5930455        cassandra:latest   "docker-entrypoint.s..." 22 hours ago   Up 5 hours (Paused)  7000-7001/tcp, 7199/tcp, 9160/tcp, 0.
0.0.0.19041->9042/tcp, :::9041->9042/tcp  lab_3_consistency_cassandra1_1
f8563a1f61c        cassandra:latest   "docker-entrypoint.s..." 22 hours ago   Up 5 hours (Paused)  7000-7001/tcp, 7199/tcp, 9160/tcp, 0.
0.0.0.19040->9042/tcp, :::9040->9042/tcp  lab_3_consistency_cassandra2_1
67d3d2d025de        cassandra:latest   "docker-entrypoint.s..." 22 hours ago   Up 5 hours (Paused)  7000-7001/tcp, 7199/tcp, 9160/tcp, 0.
0.0.0.19042->9042/tcp, :::9042->9042/tcp  lab_3_consistency_cassandra-seed_1
nayra@nayra-VirtualBox:~$
```

Left terminal window output:

```
cqlsh:bookstore> INSERT INTO books (name, category, year, title) VALUES ('Book1', 'History', 2022, 'Past Secrets');
HostUnavailable: [Unable to complete the operation against any hosts
], (msg: 227.0.0.1:9042 error: UnavailableError from server: code: 1000 [Unavailable exception] message:"Cannot achieve consistency level 1.0M" info:{'consistency': '1.0M', 'required_replicas': 3, 'all_replicas': []})
cqlsh:bookstore>
```

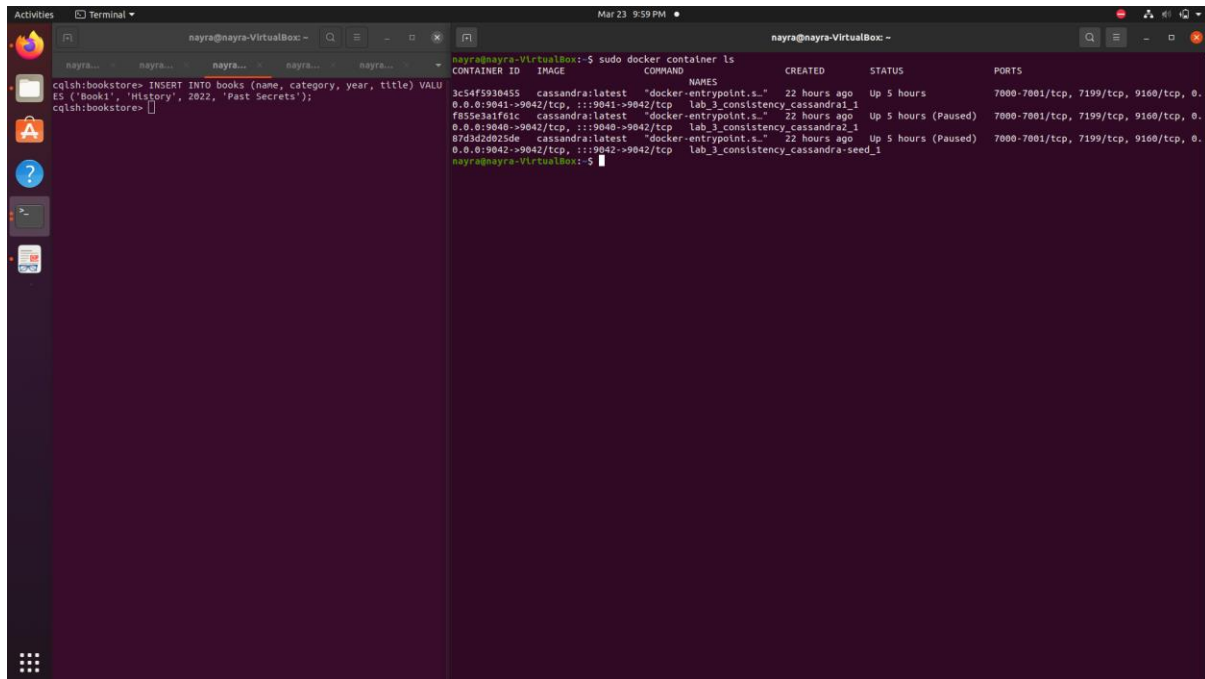
RF = 1 , Node 1&3 down

```
Activities Terminal Mar 23 9:58 PM
nayra@nayra-VirtualBox:~$ sudo docker container ls
CONTAINER ID        IMAGE               COMMAND                  CREATED        STATUS        PORTS
3c54f5930455        cassandra:latest   "docker-entrypoint.s..." 22 hours ago   Up 5 hours (Paused)  7000-7001/tcp, 7199/tcp, 9160/tcp, 0.
0.0.0.19041->9042/tcp, :::9041->9042/tcp  lab_3_consistency_cassandra1_1
f8563a1f61c        cassandra:latest   "docker-entrypoint.s..." 22 hours ago   Up 5 hours (Paused)  7000-7001/tcp, 7199/tcp, 9160/tcp, 0.
0.0.0.19040->9042/tcp, :::9040->9042/tcp  lab_3_consistency_cassandra2_1
67d3d2d025de        cassandra:latest   "docker-entrypoint.s..." 22 hours ago   Up 5 hours (Paused)  7000-7001/tcp, 7199/tcp, 9160/tcp, 0.
0.0.0.19042->9042/tcp, :::9042->9042/tcp  lab_3_consistency_cassandra-seed_1
nayra@nayra-VirtualBox:~$
```

Left terminal window output:

```
cqlsh:bookstore> INSERT INTO books (name, category, year, title) VALUES ('Book1', 'History', 2022, 'Past Secrets');
HostUnavailable: [Unable to complete the operation against any hosts
], (msg: 227.0.0.1:9042 error: UnavailableError from server: code: 1000 [Unavailable exception] message:"Cannot achieve consistency level 1.0M" info:{'consistency': '1.0M', 'required_replicas': 3, 'all_replicas': []})
cqlsh:bookstore>
```

RF = 1 , Node 2&3 down



```
nayra@nayra-VirtualBox:~$ sudo docker container ls
CONTAINER ID        IMAGE               COMMAND                  CREATED        STATUS        PORTS
3c54f593a455       cassandra:latest   "docker-entrypoint.s..." 22 hours ago   Up 5 hours    7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, :::9041->9042/tcp
f855e3a1f61c       cassandra:latest   "docker-entrypoint.s..." 22 hours ago   Up 5 hours (Paused) 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, :::9040->9042/tcp
87d3d2d925de       cassandra:latest   "docker-entrypoint.s..." 22 hours ago   Up 5 hours (Paused) 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp
nayra@nayra-VirtualBox:~$
```

Observations from first experiment (RF = 1)

- Query failed when node 1 was down -> rows (with partition key 'Book1') are stored at node 1 (cassandra1_1), so any time node 1 was down, query failed.
- Is there any way you can modify consistency to get the query to succeed if it did fail?

to increase availability we should decrease consistency and in RF = 1 , the only possible consistency level is (ANY).

ANY CL

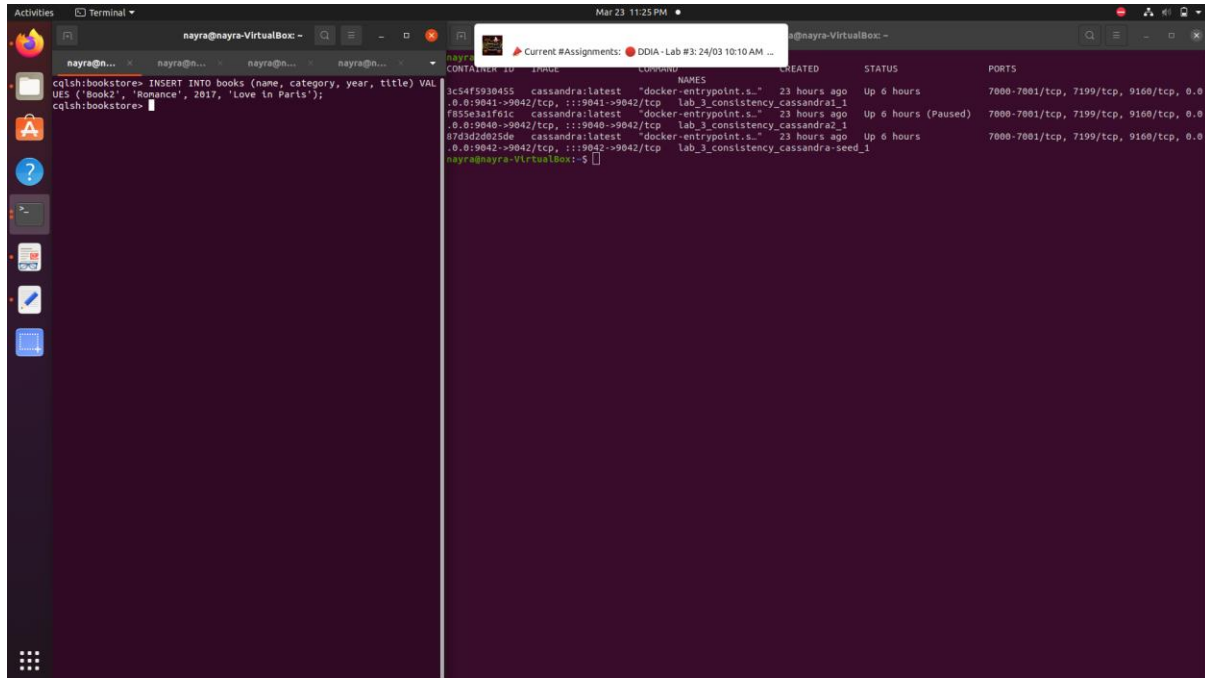
- In Cassandra, Consistency Level ANY is the weakest consistency level available. It means that a write operation is considered successful as long as at least one node (including a hinted handoff) acknowledges the write, even if the data is not immediately written to a replica.
- It is expected for any write operation to succeed with this CL as long as there is one active node.

RF = 2 , Node 1 down

The screenshot shows a terminal window with two panes. The left pane shows a cqlsh prompt where an INSERT statement was executed, followed by an error message: 'WriteTimeout: Error from server: code=100 [Coordinator node timed out waiting for replica nodes' responses] message=operation timed out ... received only 1 responses ... consistency: 'quorum', 'required_responses': 2, 'received_responses': 1, 'write_type': 'SIMPLE''. The right pane shows the output of the command 'sudo docker container ls', displaying a table of running containers.

CONTAINER ID	IMAGE	COMMAND	NAMES	CREATED	STATUS	PORTS
3c54f5930455	cassandra:latest	"docker-entrypoint.s..."		23 hours ago	Up 6 hours (Paused)	7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, :::9041->9042/tcp
f05503a1f61c	cassandra:latest	"docker-entrypoint.s..."	lab_3_consistency_cassandra1_1	23 hours ago	Up 6 hours	7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, :::9040->9042/tcp
87d3d20025de	cassandra:latest	"docker-entrypoint.s..."	lab_3_consistency_cassandra2_1	23 hours ago	Up 6 hours	7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp
			lab_3_consistency_cassandra-seed_1			

RF = 2 , Node 2 down



```
Current #Assignments: DDIA - Lab #3: 24/03 10:10 AM ...
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
3c54f5930455	cassandra:latest	"docker-entrypoint.s..."	23 hours ago	Up 6 hours	7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
.0.0:9041->9042/tcp, :::9041->9042/tcp	lab_3_consistency_cassandra1_1				
f850e3a1f61c	cassandra:latest	"docker-entrypoint.s..."	23 hours ago	Up 6 hours (Paused)	7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
.0.0:9040->9042/tcp, :::9040->9042/tcp	lab_3_consistency_cassandra2_1				
87d3d2d025de	cassandra:latest	"docker-entrypoint.s..."	23 hours ago	Up 6 hours	7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
.0.0:9042->9042/tcp, :::9042->9042/tcp	lab_3_consistency_cassandra-seed_1				

```
nayra@nayra-VirtualBox:~$
```

RF = 2 , Node 3 down

```
Activities Terminal Mar 23 11:26 PM
nayra@nayra-VirtualBox: ~
cqlsh:bookstore> INSERT INTO books (name, category, year, title) VALUES ('Book2', 'Mystery', 2010, 'Hidden Truth');
write_timeout: Error from server: code=1100 [Coordinator node timed out waiting for replica nodes' responses] message=Timeout timed out
t: received only 1 responses" info={"consistency": 'quorum', 'required_responses': 3, 'received_responses': 1, 'write_type': 'SIMPLE'}
cqlsh:bookstore>

nayra@nayra-VirtualBox:~$ sudo docker container ls
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS              PORTS
3c54f5930455        cassandra:latest   "docker-entrypoint.s" 23 hours ago        Up 6 hours          7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0.0:9041->9042/tcp, :::9041->9042/tcp   lab_3_consistency_cassandra1_1
f855e3a1f61c        cassandra:latest   "docker-entrypoint.s" 23 hours ago        Up 6 hours          7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0.0:9040->9042/tcp, :::9040->9042/tcp   lab_3_consistency_cassandra2_1
87d3d2d025de        cassandra:latest   "docker-entrypoint.s" 23 hours ago        Up 6 hours (Paused) 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0.0:9042->9042/tcp, :::9042->9042/tcp   lab_3_consistency_cassandra-seed_1
nayra@nayra-VirtualBox:~$
```

RF = 2 , Node 1&2 down

```
Activities Terminal
Mar 23 11:37 PM
nayra@naya-VirtualBox: ~
cqlsh:bookstore> INSERT INTO books (name, category, year, title) VALUES ('Book2', 'Biography', 2015, 'Life of Legends');
HostUnavailable: [Unable to complete the operation against any host
... (error details) ...]
cqlsh:bookstore>

nayra@naya-VirtualBox:~$ sudo docker container ls
CONTAINER ID        IMAGE               COMMAND                  CREATED              STATUS              PORTS
3c54f5930455        cassandra:latest   "docker-entrypoint.s..." 23 hours ago        Up 6 hours (Paused) 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0.0.0:9041->9042/tcp, :::9041->9042/tcp lab_3_consistency_cassandra1_1
f855e3a1f61c        cassandra:latest   "docker-entrypoint.s..." 23 hours ago        Up 6 hours (Paused) 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0.0.0:9040->9042/tcp, :::9040->9042/tcp lab_3_consistency_cassandra2_1
87d3d2d025de        cassandra:latest   "docker-entrypoint.s..." 23 hours ago        Up 6 hours          7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0.0.0:9042->9042/tcp, :::9042->9042/tcp lab_3_consistency_cassandra-seed_1
nayra@naya-VirtualBox:~$
```

RF = 2, Node 1&3 down

```
Activities Terminal
Mar 23 11:41 PM
nayra@naya-VirtualBox: ~
cqlsh:bookstore> INSERT INTO books (name, category, year, title) VALUES ('Book2', 'Science', 2014, 'Quantum Universe');
HostUnavailable: [Unable to complete the operation against any host
... (error details) ...]
cqlsh:bookstore>

nayra@naya-VirtualBox:~$ sudo docker container ls
CONTAINER ID        IMAGE               COMMAND                  CREATED              STATUS              PORTS
3c54f5930455        cassandra:latest   "docker-entrypoint.s..." 23 hours ago        Up 7 hours (Paused) 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0.0.0:9041->9042/tcp, :::9041->9042/tcp lab_3_consistency_cassandra1_1
f855e3a1f61c        cassandra:latest   "docker-entrypoint.s..." 23 hours ago        Up 7 hours          7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0.0.0:9040->9042/tcp, :::9040->9042/tcp lab_3_consistency_cassandra2_1
87d3d2d025de        cassandra:latest   "docker-entrypoint.s..." 23 hours ago        Up 7 hours (Paused) 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0.0.0:9042->9042/tcp, :::9042->9042/tcp lab_3_consistency_cassandra-seed_1
nayra@naya-VirtualBox:~$
```

RF = 2 , Node 2&3 down

```
Activities Terminal
nayra@nayra-VirtualBox: ~
cqlsh:bookstore> INSERT INTO books (name, category, year, title) VALUES ('Book2', 'Technology', 2013, 'AI Revolution');
Unbatched request error from server: code=1200 (Coordinator node timed out waiting for replica nodes' responses) Message="operation timed out: received only 1 responses." Info={consistency: 'QUORUM', 'request_id': 1, 'received_responses': 1, 'write_type': 'SIMPLE'}
cqlsh:bookstore>

nayra@nayra-VirtualBox:~$ sudo docker container ls
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS              PORTS
3c54f5930455       cassandra:latest   "docker-entrypoint.s" 23 hours ago        Up 6 hours          7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, :::9041->9042/tcp
f855e3a1fe1c       cassandra:latest   "docker-entrypoint.s" 23 hours ago        Up 6 hours (Paused) 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, :::9040->9042/tcp
87d3d26025d6       cassandra:latest   "docker-entrypoint.s" 23 hours ago        Up 7 hours (Paused) 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp
nayra@nayra-VirtualBox:~$
```

Observations from second experiment (RF = 2)

- Query succeeded when node 2 was down -> rows (with partition key 'Book2' aren't stored at node 2 -> so data with this partition key are hashed to node 1&3 (cassandra1_1 & cassandra_seed).
- Is there any way you can modify consistency to get the query to succeed if it did fail?

- To increase availability we should decrease consistency and in $RF = 2$, the possible consistency level is (ANY & ONE), TWO here is equivalent to QUORUM.
- Number of replicas needed with QUORUM $CL = \text{floor}(RF/2) + 1 = \text{floor}(2/2) + 1 = 2$ replicas.

ONE CL

- In Cassandra, Consistency Level ONE ensures that a write operation is acknowledged as successful when at least one replica node (not just any node) confirms the write.

ANY CL

- explained before.

Unexpected behaviors

- When node 1&3 were down, number of active replicas were expected to be 0 since the two replicas have data was down but the response indicated that active replicas = 1.

Possible reasons

1. Cassandra uses a gossip protocol to detect node failures. If one replica node has been marked as down (UNREACHABLE), but the other has not yet been fully detected as down, the coordinator node may only report one unavailable replica. This happens because failure detection in Cassandra is eventually consistent—nodes learn about failures at different times.
2. Cassandra uses a gossip protocol to detect node failures. If one replica node has been marked as down (UNREACHABLE), but the other has not yet been fully detected as down, the coordinator node may only report one unavailable replica. This happens because failure detection in Cassandra is eventually consistent—nodes learn about failures at different times.

3. If hinted handoff is enabled, Cassandra might temporarily assume that one of the replicas can still receive writes later.
- Insert query succeeded only one time and while selecting data in table there are 4 rows were inserted.

Possible reasons :

1. If the write fails, Cassandra may still log the data in the commit log of available nodes, or hints may be stored to be replayed later.

Expected behavior with decreasing CL

Nodes Down	RF = 2 (ONE)	RF = 2 (ANY)
Node 1 down	OK	OK
Node 2 down	OK	OK
Node 3 down	OK	OK
Node 1&2 down	OK	OK
Node 1&3 down	ERROR	OK
Node 2&3 down	OK	OK

ANY CL

- It is expected for any write operation to succeed with this CL as long as there is one active node.

ONE CL

- since data is stored at node 1&3 , query will fail only when node 1&3 are down together and succeed otherwise.

```
Activities Terminal
nayra@nayra-VirtualBox: ~
nayra@n... x nayra@n... x nayra@n... x nayra@n...
cqlsh:bookstore> select * from books where name = 'Book2';
name | category | year | title
-----
Book2 | Mystery | 2016 | Hidden Truth
Book2 | Romance | 2017 | Love in Paris
Book2 | Technology | 2013 | AI Revolution
Book2 | Thriller | 2018 | Silent Chase
(4 rows)
cqlsh:bookstore> select * from books;
name | category | year | title
-----
Book1 | Fiction | 2024 | The Mystery
Book1 | History | 2022 | Past Secrets
Book1 | Sci-Fi | 2023 | Future World
Book2 | Mystery | 2016 | Hidden Truth
Book2 | Romance | 2017 | Love in Paris
Book2 | Technology | 2013 | AI Revolution
Book2 | Thriller | 2018 | Silent Chase
(7 rows)
cqlsh:bookstore>
```

```
nayra@nayra-VirtualBox:~$ sudo docker container ls
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
3c54f5930455   cassandra:latest   "docker-entrypoint.s..." 23 hours ago   Up 7 hours    7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, :::9041->9042/tcp
f855e3a1f61c   cassandra:latest   "docker-entrypoint.s..." 23 hours ago   Up 7 hours    7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, :::9040->9042/tcp
87d3d2d025de   cassandra:latest   "docker-entrypoint.s..." 23 hours ago   Up 7 hours    7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp
nayra@nayra-VirtualBox:~$
```

RF = 3, Node 1 down

```
Activities Terminal
nayra@nayra-VirtualBox: ~
nayra@n... x nayra@n... x nayra@n... x nayra@n...
cqlsh:bookstore> INSERT INTO books (name, category, year, title) VALUES ('Book3', 'Poetry', 2012, 'Echoes of Time');
cqlsh:bookstore>
```

```
nayra@nayra-VirtualBox:~$ sudo docker container ls
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
3c54f5930455   cassandra:latest   "docker-entrypoint.s..." 24 hours ago   Up 7 hours (Paused)    7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, :::9041->9042/tcp
f855e3a1f61c   cassandra:latest   "docker-entrypoint.s..." 24 hours ago   Up 7 hours    7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, :::9040->9042/tcp
87d3d2d025de   cassandra:latest   "docker-entrypoint.s..." 24 hours ago   Up 7 hours    7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp
nayra@nayra-VirtualBox:~$
```

RF = 3, Node 2 down

```
Activities Terminal
Mar 24 12:31 AM
nayra@nayra-VirtualBox: ~
cqlsh:bookstore> INSERT INTO books (name, category, year, title) VALUES ('Books', 'Drama', 2011, 'The Last Act');
cqlsh:bookstore>

nayra@nayra-VirtualBox:~$ sudo docker container ls
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS              PORTS
3c54f5930455        cassandra:latest   "docker-entrypoint.s" 24 hours ago        Up 7 hours          7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0:9041->9042/tcp, :::9041->9042/tcp    lab_3_consistency_cassandra1_1
f855e3a1f61c        cassandra:latest   "docker-entrypoint.s" 24 hours ago        Up 7 hours (Paused) 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0:9040->9042/tcp, :::9040->9042/tcp    lab_3_consistency_cassandra2_1
87d3d2d025de        cassandra:latest   "docker-entrypoint.s" 24 hours ago        Up 7 hours          7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0:9042->9042/tcp, :::9042->9042/tcp    lab_3_consistency_cassandra-seed_1
nayra@nayra-VirtualBox:~$
```

RF = 3 , Node 3 down

```
Activities Terminal
nayra@nayra-VirtualBox: ~
cqlsh:bookstore> INSERT INTO books (name, category, year, title) VALUES ('Books', 'Philosophy', 2010, 'The Thinker's Path');
cqlsh:bookstore>

nayra@nayra-VirtualBox:~$ sudo docker container ls
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS              PORTS
3c54f5930455       cassandra:latest   "docker-entrypoint.s" 24 hours ago       Up 7 hours         7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0:9041->9042/tcp, :::9041->9042/tcp   lab_3_consistency_cassandra1_1
f855e3a1f61c       cassandra:latest   "docker-entrypoint.s" 24 hours ago       Up 7 hours         7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0:9040->9042/tcp, :::9040->9042/tcp   lab_3_consistency_cassandra2_1
87d3d2d025de       cassandra:latest   "docker-entrypoint.s" 24 hours ago       Up 7 hours (Paused) 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0:9042->9042/tcp, :::9042->9042/tcp   lab_3_consistency_cassandra-seed_1
nayra@nayra-VirtualBox:~$
```

RF = 3 , Node 1&2 down

```
Activities Terminal Mar 24 12:34 AM
nayra@nayra-VirtualBox: ~
cqlsh:bookstore> INSERT INTO books (name, category, year, title) VALUES ('Books', 'Self-Help', 2009, 'Mindset Matters');
HostUnavailable: [Unable to complete the operation against any host
... (message) ...]
cqlsh:bookstore>

nayra@nayra-VirtualBox:~$ sudo docker container ls
CONTAINER ID        IMAGE               COMMAND                  CREATED              STATUS              PORTS
3c54f5930455        cassandra:latest   "docker-entrypoint.s..." 24 hours ago        Up 7 hours (Paused)  7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0.0:9041->9042/tcp, :::9041->9042/tcp    lab_3_consistency_cassandra1
f855e3a1f61c        cassandra:latest   "docker-entrypoint.s..." 24 hours ago        Up 7 hours (Paused)  7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0.0:9040->9042/tcp, :::9040->9042/tcp    lab_3_consistency_cassandra2_1
87d3d2d025de        cassandra:latest   "docker-entrypoint.s..." 24 hours ago        Up 7 hours          7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0.0:9042->9042/tcp, :::9042->9042/tcp    lab_3_consistency_cassandra-seed_1
nayra@nayra-VirtualBox:~$
```

RF = 3 , Node 1&3 down

```
Activities Terminal Mar 24 12:35 AM
nayra@nayra-VirtualBox: ~
cqlsh:bookstore> INSERT INTO books (name, category, year, title) VALUES ('Books', 'Comics', 2008, 'Superhero Origins');
write_timeout: Error from server: code=1100 [Coordinator node timed out waiting for replica nodes' responses] message=Timeout timed out
t: received only 1 responses" info={"consistency": 'quorum', 'required_responses': 3, 'received_responses': 1, 'write_type': 'SIMPLE'}
cqlsh:bookstore>

nayra@nayra-VirtualBox:~$ sudo docker container ls
CONTAINER ID        IMAGE               COMMAND                  CREATED              STATUS              PORTS
3c54f5930455        cassandra:latest   "docker-entrypoint.s..." 24 hours ago        Up 7 hours (Paused)  7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0.0:9041->9042/tcp, :::9041->9042/tcp  lab_3_consistency_cassandra1_1
f855e3a1f61c        cassandra:latest   "docker-entrypoint.s..." 24 hours ago        Up 7 hours          7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0.0:9040->9042/tcp, :::9040->9042/tcp  lab_3_consistency_cassandra2_1
87d3d2d025de        cassandra:latest   "docker-entrypoint.s..." 24 hours ago        Up 7 hours (Paused)  7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0
0.0.0:9042->9042/tcp, :::9042->9042/tcp  lab_3_consistency_cassandra-seed_1
nayra@nayra-VirtualBox:~$
```

RF = 3 , Node 2&3 down

```

cqlsh:bookstore> INSERT INTO books (name, category, year, title) VALUES ('Books', 'Travel', 2007, 'Around the World');
UnavailableException: Unable to complete the operation against any host.
[Error: [127.0.0.1]:9042 socket: unavailable (Error from server: code=9000 [UnavailableException] message='Cannot achieve consistency level QUORUM' info={'consistency': 'QUORUM', 'required_replicas': 3, 'native_transport_id': 1})]]
cqlsh:bookstore>

nayra@nayra-VirtualBox:~$ sudo docker container ls
CONTAINER ID        IMAGE               COMMAND                  CREATED              STATUS              PORTS
3c54f5930455        cassandra:latest   "docker-entrypoint.s..." 24 hours ago        Up 7 hours         7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, :::9041->9042/tcp
f855e3a1f61c        cassandra:latest   "docker-entrypoint.s..." 24 hours ago        Up 7 hours (Paused) 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, :::9040->9042/tcp
87d3d2d025de        cassandra:latest   "docker-entrypoint.s..." 24 hours ago        Up 7 hours (Paused) 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp
nayra@nayra-VirtualBox:~$

```

Is there any way you can modify consistency to get the query to succeed if it did fail?

- To increase availability we should decrease consistency and in RF = 3 , the possible consistency level is (ANY & ONE) , TWO here is equivalent to QUORUM.
- Number of replicas needed with QUORUM CL = floor (RF/2) + 1 = floor (2/2)+1 = 2 replicas.

ONE CL & ANY CL

- explained before.

Unexpected behaviors

- Insert query succeeded only one time and while selecting data in table there are 4 rows were inserted.

Possible reasons :

1. If the write fails, Cassandra may still log the data in the commit log of available nodes, or hints may be stored to be replayed later.

The screenshot shows a terminal window with two panes. The left pane shows a SQL query being executed in a database, resulting in a table with 4 rows. The right pane shows the output of a Docker command listing containers.

```
cqlsh:bookstore> select * from books where name = 'Book3' ;
```

name	category	year	title
Book3	Comics	2008	Superhero Origins
Book3	Drama	2011	The Last Act
Book3	Philosophy	2010	The Thinker's Path
Book3	Poetry	2012	Echoes of Time

(4 rows)
cqlsh:bookstore>

```
nayra@nayra-VirtualBox:~$ sudo docker container ls
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
3c54f5930455	cassandra:latest	"docker-entrypoint.s..."	24 hours ago	Up 7 hours	7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9041->9042/tcp, :::9041->9042/tcp
f855e3a1f61c	cassandra:latest	"docker-entrypoint.s..."	24 hours ago	Up 7 hours	7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9040->9042/tcp, :::9040->9042/tcp
87d3d2d825de	cassandra:latest	"docker-entrypoint.s..."	24 hours ago	Up 7 hours	7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, :::9042->9042/tcp

Expected behavior with decreasing CL

Nodes Down	RF = 3 (ONE)	RF = 3 (ANY)
Node 1 down	OK	OK
Node 2 down	OK	OK
Node 3 down	OK	OK
Node 1&2 down	OK	OK
Node 1&3 down	OK	OK
Node 2&3 down	OK	OK

ANY CL

- It is expected for any write operation to succeed with this CL as long as there is one active node.

ONE CL

- since data is stored at node 1&3all nodes , query will succeed as long as there is one active node.